

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	AIDS / 2024	Date:	17/08/2026

Code of Conduct

I M.KASIF JALAL (192424214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: kasif jalal M

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:

Incorrect interpretation of ambiguous sentences

Failure to handle subject–verb agreement

Inefficient parsing for long queries

Example user query:

“Show me the transactions with the card from last month”

a) Analyze the ambiguity present in the given sentence using CFG

b) Evaluate the limitations of CFG in handling real-world conversational queries

c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing

d) Justify how your proposed method improves accuracy and efficiency in an industry setting

Program:

```
sentence="The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."
```

```
tokens=sentence.replace(".", "").split()
```

```
print("HEALTHCARE SEMANTIC ANALYSIS")
```

```
print("\nInput:")
```

```
print(sentence)
```

```
print("\nTokens:")
```

```
print(tokens)
```

```
print("\nSyntactic Structure:")
```

```
print("Doctor -> recommends -> starting medication")
```

```
print("Doctor -> recommends -> scheduling follow-up visit")
```

```
print("who -> reviewed -> patient")
```

```
print("follow-up visit -> in -> Chennai")
```

```
print("\nSemantic Roles:")
```

```
print("Doctor -> Agent")
```

```
print("Patient -> Patient Entity")
```

```
print("Medication -> Treatment")
```

```
print("Starting medication -> Treatment Action")
```

```
print("Follow-up visit -> Follow-up Action")
```

```
print("Chennai -> Location")
```

```
print("\nSemantic Representation:")
```

```
print("DOCTOR(x)")
```

```
print("REVIEWS(x,PATIENT)")
```

```
print("TIME(Review,LastWeek)")
```

```
print("RECOMMENDS(x,StartMedication)")
```

```
print("RECOMMENDS(x,ScheduleFollowUp)")
```

```
print("LOCATION(FollowUp,Chennai)")
```

```
print("\nStructured Output:")
```

```
print("Diagnosis: Not explicitly stated")
```

```
print("Treatment: Start medication")
```

```
print("Follow-up: Schedule follow-up visit")
```

```
print("Location: Chennai")
```

```
print("Review Time: Last week")
```

```
print("\nArchitecture:")
```

```
print("CFG -> Earley Parser -> PCFG -> Feature Structures ->")
```

```
print("Sub-Categorization -> Semantic Role Labeling -> Structured Output")
```

Output:

2.

3. HEALTHCARE SEMANTIC ANALYSIS

4.

5. Input:

6. The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai.

7.

8. Tokens:

9. ['The', 'doctor', 'who', 'reviewed', 'the', 'patient', 'last', 'week', 'recommends', 'starting', 'medication', 'and', 'scheduling', 'a', 'follow-up', 'visit', 'in', 'Chennai']

10.

11. Syntactic Structure:

12. Doctor -> recommends -> starting medication

13. Doctor -> recommends -> scheduling follow-up visit

14. who -> reviewed -> patient

15. follow-up visit -> in -> Chennai

16.

17. Semantic Roles:

18. Doctor -> Agent

19. Patient -> Patient Entity

20. Medication -> Treatment

21. Starting medication -> Treatment Action

22. Follow-up visit -> Follow-up Action

23. Chennai -> Location

24.

25. Semantic Representation:

26. DOCTOR(x)

27. REVIEWS(x, PATIENT)

28. TIME(Review, LastWeek)

29. RECOMMENDS(x, StartMedication)

30. RECOMMENDS(x, ScheduleFollowUp)

31. LOCATION(FollowUp, Chennai)

32.

33. Structured Output:

34. Diagnosis: Not explicitly stated

35. Treatment: Start medication

36. Follow-up: Schedule follow-up visit

37. Location: Chennai

38. Review Time: Last week

39.

40. Architecture:

41. CFG -> Earley Parser -> PCFG -> Feature structures ->

42. Sub-Categorization -> Semantic Role Labeling -> Structured Output
A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:

Backtracking issues

Poor handling of incomplete or ambiguous input

Difficulty in real-time response

Example command:

“Book a flight to Delhi with a window seat”

a) Analyze how the given sentence can lead to multiple parse structures

b) Analyze the limitations of top-down parsing in this real-time application

c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs

d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

Program:

command="Book a flight to Delhi with a window seat"

print("VOICE ASSISTANT ANALYSIS")

```

print("Input:",command)

print("\nPossible Parse 1:")
print("BOOK(FLIGHT, DESTINATION=Delhi, SEAT=Window)")

print("\nPossible Parse 2:")
print("BOOK(FLIGHT(Delhi), WITH(WindowSeat))")

print("\nSelected Interpretation:")
print("BOOK(Flight, Destination=Delhi, Seat=Window)")

print("\nTop-Down Parsing:")
print("May require backtracking for ambiguous structures")

print("\nEarley Parsing:")
print("Maintains partial parse states and handles ambiguity efficiently")

```

Output:

VOICE ASSISTANT ANALYSIS

Input: Book a flight to Delhi with a window seat

Possible Parse 1:

BOOK(FLIGHT, DESTINATION=Delhi, SEAT=Window)

Possible Parse 2:

BOOK(FLIGHT(Delhi), WITH(WindowSeat))

Selected Interpretation:

BOOK(Flight, Destination=Delhi, Seat=Window)

Top-Down Parsing:

May require backtracking for ambiguous structures

Earley Parsing:

Maintains partial parse states and handles ambiguity efficiently

43. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.

The system must handle:

Complex and lengthy medical sentences

Ambiguity in sentence structure

Accurate subject-verb agreement

Relationships between medical actions and entities (sub-categorization)

Real-time processing for hospital use

Example input:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.

a) Design a complete NLP architecture integrating:

CFG for syntactic structure

Probabilistic CFG (PCFG) for ambiguity resolution

Feature Structures for agreement handling

Efficient parsing techniques

b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location).

c) Create a method to:

Incorporate sub-categorization frames for medical verbs

Handle real-time data processing

Improve accuracy and scalability in a hospital environment

Program:

```
from collections import defaultdict
```

```
rules={
    "S":["VP",1.0],
    "VP":["V NP",1.0],
    "NP":["Det N PP",0.6],["Det N",0.4],
    "PP":["P NP",1.0]
}
```

```
query="Show me the transactions with the card from last month"
```

```
print("BANKING QUERY ANALYSIS")
print("Input:",query)
print("\nPossible Interpretations:")
print("1. Transactions made with the card during last month")
print("2. Transactions associated with a particular card from last month")
print("\nMost Likely Semantic Interpretation:")
print("QUERY(Transactions, Card, Time=LastMonth)")
print("\nParsing Method: Earley Parser")
print("Ambiguity Resolution: PCFG")
print("Agreement Handling: Feature Structures")
```

Output:

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____